

AWS ARCHITECTURE

NutriTrack

Cloud Infrastructure Design

Enterprise-Grade AWS Infrastructure on us-east-1 | High Availability | Secure by Design

Region: us-east-1

Multi-AZ: 2 AZs

VPC: 10.0.0.0/16



Slide 2 — Architecture Overview & Design Principles

Design Principles

- High Availability — Dual-AZ deployment (us-east-1a / us-east-1b) eliminates single points of failure across all tiers.
- Defense-in-Depth — Strict Security Group chain (ALB → Frontend → Internal ALB → Backend → DB/Cache); no direct public access to private resources.
- Elastic Scalability — Auto Scaling Groups on Frontend and Backend tiers automatically respond to traffic demand without manual intervention.
- Operational Simplicity — Managed services (RDS Multi-AZ) reduce undifferentiated heavy lifting; Bastion Host centralizes SSH access control.



VPC

10.0.0.0/16



Tiers

3-Tier Architecture



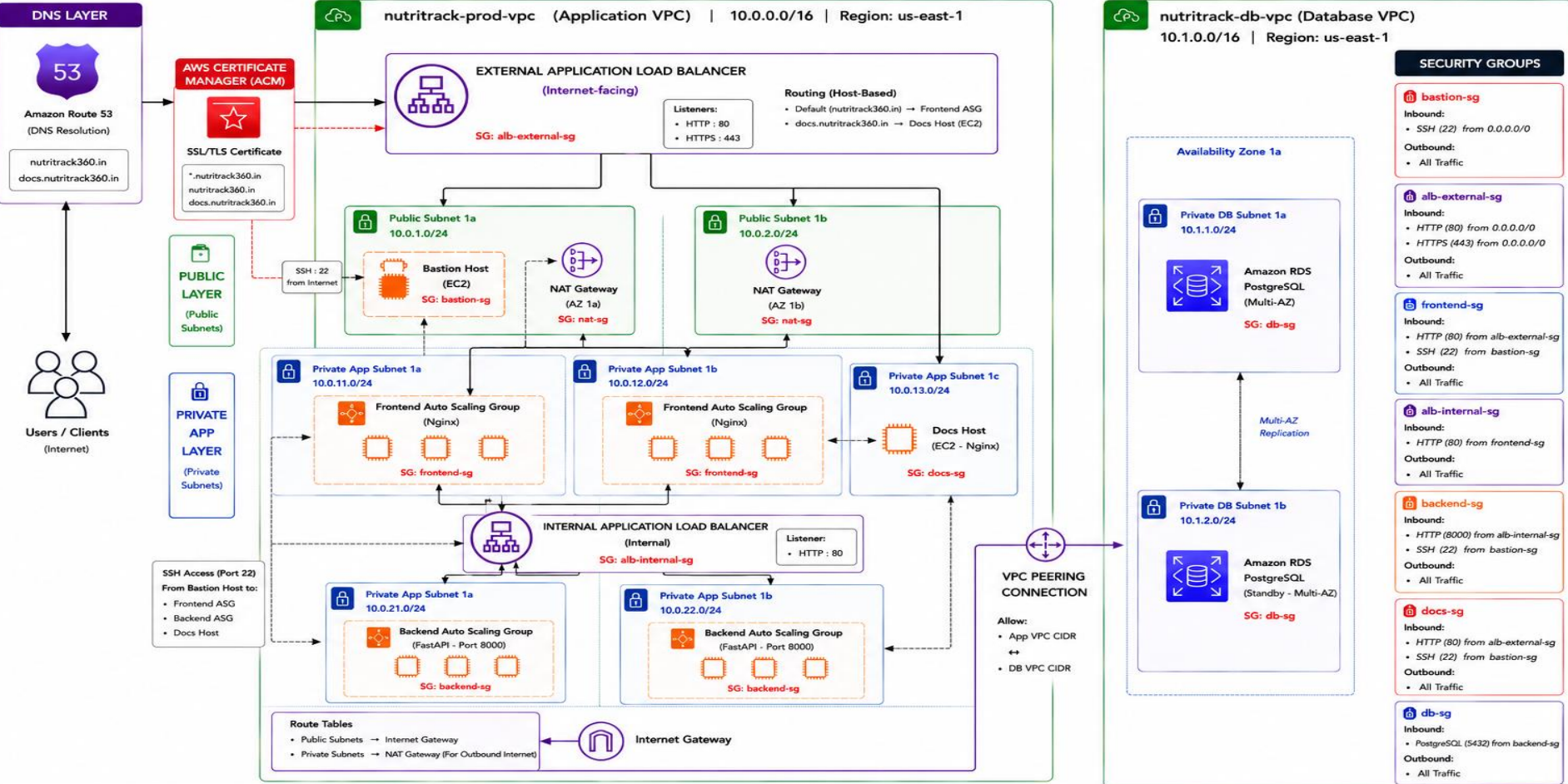
Compute

ASG + EC2 + Bastion



Data

RDS PostgreSQL Multi-AZ



LEGEND

- ALB
- EC2 Instance
- Auto Scaling Group
- NAT Gateway
- RDS (Multi-AZ)
- Public Subnet
- Private App Subnet
- Private DB Subnet

FLOW SUMMARY

- User → Route 53 → External ALB (ACM SSL)
- External ALB → Frontend ASG (default) / Docs Host (host-based)
- Frontend ASG → Internal ALB → Backend ASG
- Backend ASG → RDS PostgreSQL (via VPC Peering)
- Bastion Host → SSH to Frontend / Backend / Docs
- Private Subnets → NAT Gateways (Outbound Internet)

Slide 3 — VPC Design & Network Topology

VPC & Subnet Architecture

- Single VPC (10.0.0.0/16) spans us-east-1a and us-east-1b, providing a private, isolated network boundary for all NutriTrack resources.
- Six subnets across three tiers: Public (10.0.1.0/24, 10.0.2.0/24), Private App (10.0.11.0/24, 10.0.12.0/24), and Private DB (10.0.21.0/24, 10.0.22.0/24).
- Tier isolation enforced via separate subnet routing tables — DB subnets have no route to the internet, preventing accidental data exposure.
- Redundant NAT Gateways (one per AZ) in public subnets provide fault-tolerant outbound internet access for private instances without inbound exposure.

VPC 10.0.0.0/16

PUBLIC SUBNETS

10.0.1.0/24 | 10.0.2.0/24

us-east-1a

us-east-1b

PRIVATE APP SUBNETS

10.0.11.0/24 | 10.0.12.0/24

us-east-1a

us-east-1b

PRIVATE DB SUBNETS

10.0.21.0/24 | 10.0.22.0/24

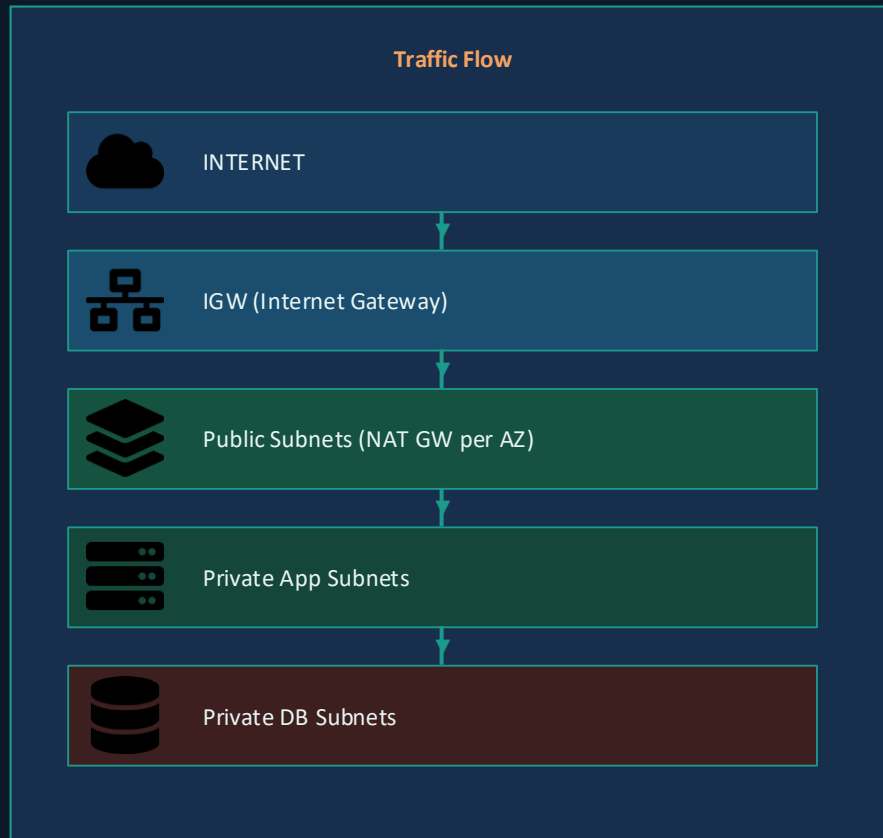
us-east-1a

us-east-1b

Slide 4 — Internet & NAT Gateway Strategy

Ingress & Egress Routing

- Internet Gateway (IGW) attached to the VPC enables bidirectional internet traffic exclusively for resources in the two Public Subnets.
- Redundant NAT Gateways — one deployed in each public subnet AZ — ensure private instances retain outbound internet access even if one AZ fails.
- Private subnet route tables direct all 0.0.0.0/0 egress to the AZ-local NAT Gateway, minimizing cross-AZ data transfer costs and latency.
- No inbound internet traffic is permitted directly to private App or DB subnets; all ingress flows exclusively through the External ALB.

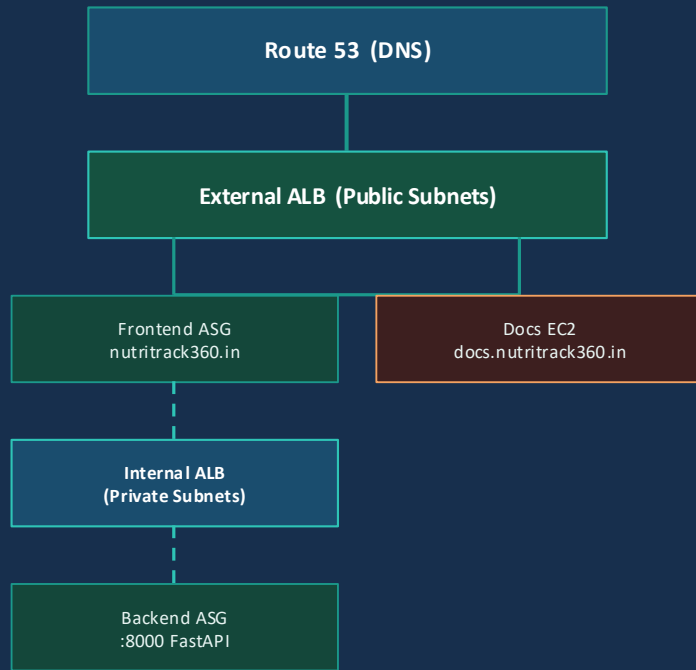


Slide 5 — DNS, Load Balancing & Routing Rules

Load Balancer Architecture

- Route 53 handles DNS resolution for `nutrtrack360.in` and `docs.nutrtrack360.in`, routing all traffic into the External ALB in the public subnets.
- External ALB performs host-based routing: default requests (`nutrtrack360.in`) are forwarded to the Frontend ASG; `docs.nutrtrack360.in` routes to the standalone Docs EC2 instance.
- Internal ALB (in private app subnets) handles east-west traffic, routing API calls from Frontend EC2 instances to the Backend ASG on Port 8000 — never exposed to the internet.
- ALB health checks continuously validate target availability and automatically deregister unhealthy instances from the target group.

Routing Diagram



Slide 6 — Compute Tier: Frontend, Backend & Docs



Frontend ASG

- EC2 instances running Nginx in native config
- Auto Scaling Group across 2 AZs
- Receives HTTP(S) from External ALB
- Serves SPA and proxies API → Internal ALB



Backend ASG

- EC2 instances running FastAPI on Port 8000
- Auto Scaling Group across 2 AZs
- Accessible only via Internal ALB
- Connects to RDS PostgreSQL over private subnet



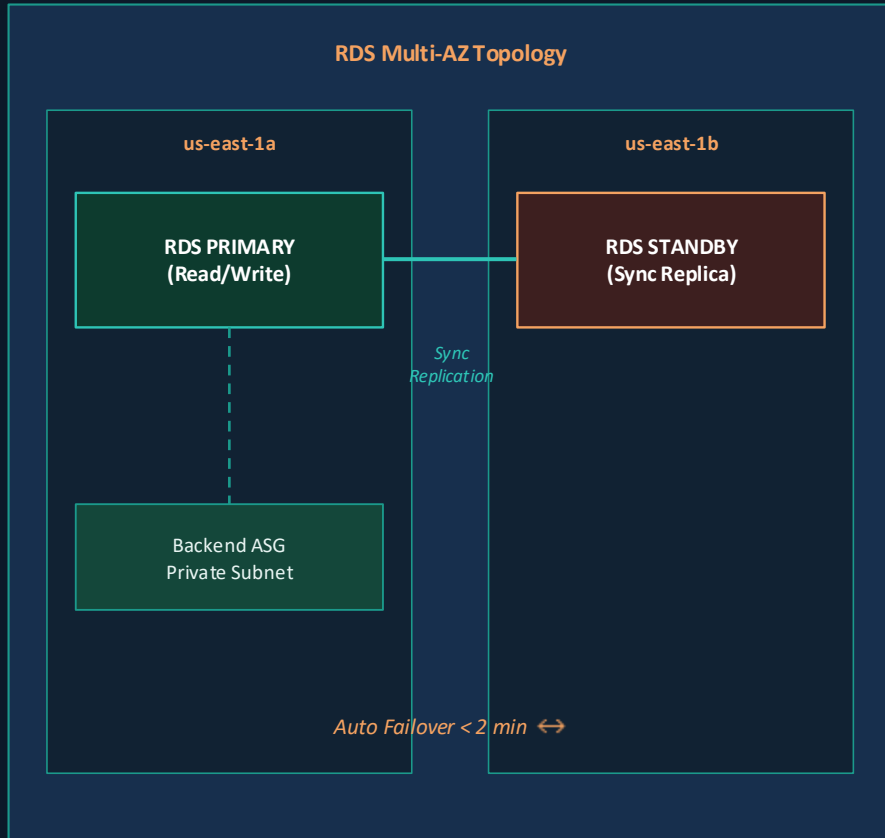
Docs + Bastion

- Single Ubuntu EC2 serving docs via Nginx
- Not part of ASG — static single instance
- Bastion Host in public subnet for SSH access
- SSH (Port 22) only via Bastion SG → private subnets

Slide 7 — Data & Caching Tier

Database Architecture

- Amazon RDS PostgreSQL deployed in Multi-AZ mode — a standby replica in a second AZ provides automatic failover with zero data loss in the event of a primary instance failure.
- Database instances reside exclusively in the Private DB Subnets (10.0.21.0/24, 10.0.22.0/24) with no internet route, accessible only to Backend ASG instances via Security Group rules.
- Multi-AZ deployment ensures RPO ≈ 0 (synchronous replication) and RTO < 2 minutes (automated promotion), meeting enterprise-grade availability SLAs.
- Subnet Group spans both AZs, enabling RDS to place primary and standby in geographically isolated failure domains.



Slide 8 — Security Architecture & Group Chain

Security Group Model

- All ingress is governed by a strict Security Group chain: External ALB SG → Frontend SG → Internal ALB SG → Backend SG → RDS SG. Each layer accepts traffic only from the immediately preceding SG.
- SSH access (Port 22) to all private EC2 instances is permitted exclusively from the Bastion Host Security Group, eliminating direct internet SSH exposure.
- The External ALB SG is the only resource accepting inbound internet traffic (Ports 80/443). All other Security Groups have no direct inbound rules from 0.0.0.0/0.
- RDS Security Group accepts connections only on Port 5432 from the Backend ASG SG — completely isolated from the Frontend tier and the public internet.

Security Group Chain



Slide 9 — High Availability & Resilience Summary



Dual-AZ VPC

All subnet tiers replicated across us-east-1a & us-east-1b — no single AZ failure can take down the service.



ASG Compute

Frontend & Backend Auto Scaling Groups automatically replace failed instances and scale on demand.



Dual NAT GW

One NAT Gateway per AZ — private outbound traffic is never dependent on a single NAT Gateway.



RDS Multi-AZ

Synchronous standby replica with < 2 min automated failover. RPO $\approx$ 0, RTO < 2 minutes.



Dual ALB

Both External and Internal ALBs span multiple AZs and perform continuous health-check-based routing.



SG Isolation

Each tier is isolated by Security Group — a failure or breach in one tier cannot directly propagate to others.

ARCHITECTURE SUMMARY

NutriTrack AWS Infrastructure

A Resilient, Secure, and Scalable Three-Tier Cloud Architecture

Layer	Technology	HA Pattern
Networking	VPC 10.0.0.0/16, 6 Subnets	Dual-AZ, Dual NAT GW
DNS & Routing	Route 53 + External ALB	Multi-AZ ALB, Host-Based Rules
Frontend Compute	EC2 ASG + Nginx	Auto Scaling across 2 AZs
Backend Compute	EC2 ASG + FastAPI :8000	Auto Scaling across 2 AZs
Documentation	Ubuntu EC2 + Nginx	Single Instance (Docs)
Database	RDS PostgreSQL Multi-AZ	Sync Replica, < 2min RTO
Security	Security Groups (chain model)	Zero Trust Ingress Chain